

KUSHAL SWARGAM

AI Developer | Full-Stack Engineer

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SUMMARY

First-year B.Tech Computer Science student specializing in applied AI — building local-first LLM applications, computer-vision systems, and retrieval-augmented memory architectures. Hands-on experience integrating open-source LLMs (Ollama / Gemma / Qwen), vector databases, speech pipelines, and agentic tools, paired with full-stack engineering skills to ship complete products end-to-end. Seeking an AI Developer internship to build production-grade, real-world AI systems.

TECHNICAL SKILLS

AI / ML: Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Embeddings, OpenCV, MediaPipe, ChromaDB (Vector DB), Whisper (STT), Ollama, Local LLM Deployment, Gemma / Qwen LLMs, Sentiment Analysis

Backend: Python, FastAPI, Node.js, REST APIs, Electron, Spoonacular API

Frontend: React 18, Vite, Tailwind CSS, Three.js, Zustand, TanStack Query, React Router, shadcn/ui

Languages: Python, JavaScript, C, C++, HTML5, CSS3

Tools & Platforms: Git, GitHub, npm, pip, VS Code, @dnd-kit, Recharts

PROJECTS

Aura — AI Desktop Companion — github.com/kushal-sw/ai-companion

React, Electron, FastAPI, Python, Three.js, ChromaDB, Ollama (Gemma 3:4b), Whisper

- Built a privacy-first AI desktop assistant with a real-time animated 3D VRM avatar, voice interaction, and a retrieval-augmented memory system using ChromaDB vector embeddings for persistent, context-aware conversations.
- Engineered prompt pipelines for a locally-hosted Ollama LLM (Gemma 3:4b); integrated OpenAI Whisper for speech-to-text and Coqui TTS for expressive voice responses, with real-time sentiment analysis driving avatar expressions.
- Architected a full-stack Electron + React frontend communicating with a FastAPI backend — 100% on-device inference with no cloud dependency, directly addressing data-privacy constraints common in production AI deployments.

DevDNA — Agentic Local AI Code Assistant — github.com/kushal-sw/dev-dna

Python, FastAPI, Ollama (Qwen3:8b), TypeScript, Shell

- Developed a fully local, agentic CLI coding assistant powered by Ollama (Qwen3:8b) that autonomously reads, writes, and modifies files, with colored diff previews before applying any changes.
- Implemented adaptive prompt strategies (Beginner / Intermediate / Advanced) with real-time pattern detection for naming conventions and code style, improving LLM output relevance to developer preference.
- Designed for 100% on-device privacy with no API keys or cloud calls — configurable across multiple local models (Qwen3, CodeLlama, DeepSeek-Coder).

Hand Gesture Mouse Control — github.com/kushal-sw/hand-gesture-mouse-control

Python, OpenCV, MediaPipe, PyAutoGUI

- Developed a computer-vision application that maps 21 MediaPipe hand landmarks to real-time mouse control, with configurable sensitivity and gesture recognition for click, right-click, and stop actions.

Recipe Discovery App — github.com/kushal-sw/react-mini-project

React 18, Vite, Tailwind CSS, Zustand, TanStack Query, React Router v6, Recharts, @dnd-kit, Spoonacular API

- Built a full-featured recipe discovery web app with search by dish/ingredient, nutrition panels, dietary filters, dark mode, and a drag-and-drop weekly meal planner with auto-generated shopping lists.
- Managed global state with Zustand (persist middleware) and reduced redundant network requests via TanStack Query caching of Spoonacular API responses.

Browser History CLI — Linked List Implementation — github.com/kushal-sw/browser-cli-linkedlist

C++

- Implemented a command-line browser history simulator in C++ using a doubly linked list to support back/forward navigation, history persistence, and animated terminal output; forked by peers in the community.

EDUCATION

B.Tech in Computer Science & Engineering

Expected Graduation: 2028 | Currently: 1st Year

ACHIEVEMENTS & ACTIVITIES

- Hackathon Finalist — shortlisted among top teams for ability to build and pitch under time constraints.
- Built and maintained 5+ featured end-to-end projects in 1st year spanning agentic AI, local LLM tooling, computer vision, and full-stack web — version-controlled across 26 public GitHub repositories. Self-driven learner who independently explored vector databases, retrieval-augmented architectures, and 3D rendering (Three.js/VRM).